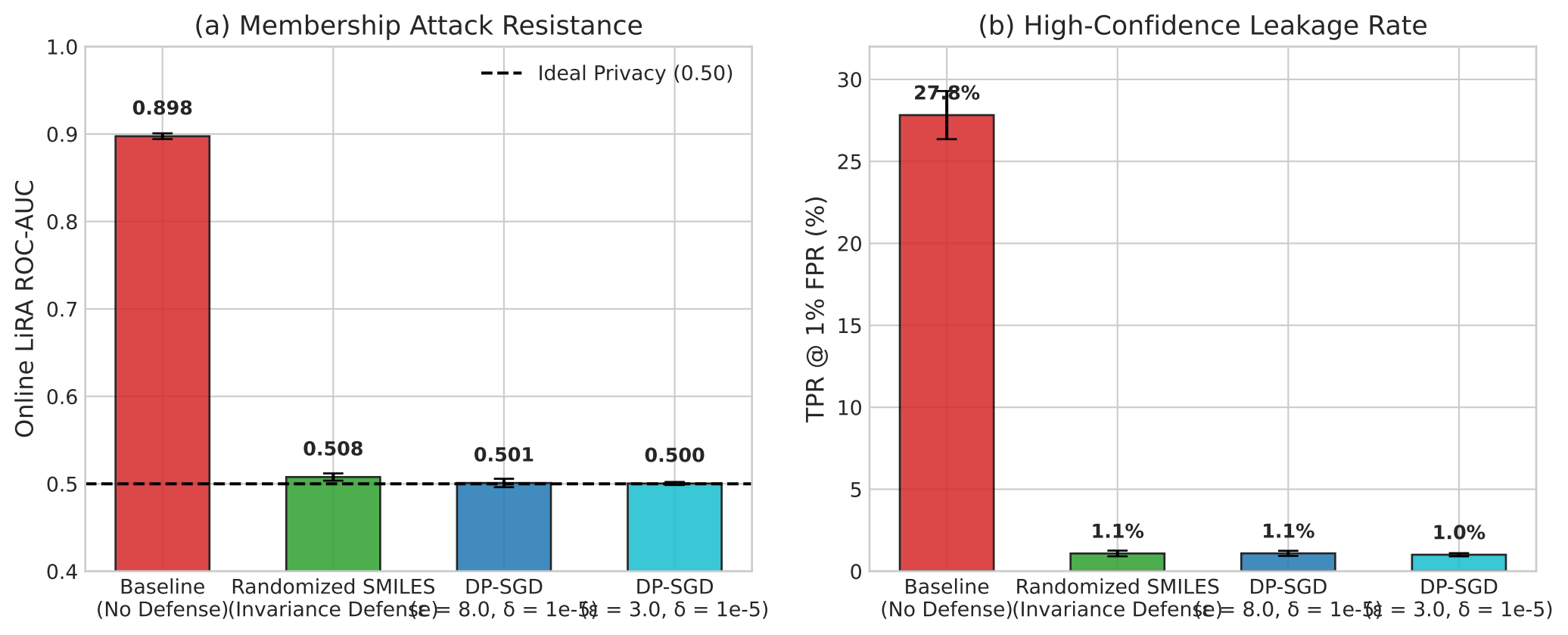
(a) Membership Attack Resistance
Online LiRA ROC-AUC
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Ideal Privacy (0.50)
0.898
0.508
0.501
0.500
Baseline (No Defense)
Randomized SMILES (Invariance Defense)
DP-SGD (ε = 8.0, δ = 1e-5)
DP-SGD (ε = 3.0, δ = 1e-5)
(b) High-Confidence Leakage Rate
TPR @ 1% FPR (%)
30
25
20
15
10
5
0
27.8%
1.1%
1.1%
1.0%
Baseline (No Defense)
Randomized SMILES (Invariance Defense)
DP-SGD (ε = 8.0, δ = 1e-5)
DP-SGD (ε = 3.0, δ = 1e-5)